The amplitude model as a generator

What `amp2609` is, where it may be used, and what it inherits

V. Kubarovsky
Jefferson Lab

18 September 2026

Abstract

`exclurad_py` now generates π^0 and η electroproduction from a model fitted directly to the world data at the level of the helicity amplitudes. This note covers the join between the two: which parameter vector is installed, what cross section is sampled, how far the sampling box reaches beyond the region the model was fitted in, whether the sampled cross section stays positive, and in what sense the radiative-correction loop is and is not closed. The physics of the model itself is not repeated here; it is in `~/rc_iter1/tex/report.pdf`, and the radiative corrections are in `doc/note/exclurad_py_note.pdf`. This note is only the seam.

1 What is installed

registered name	`pi0.amp2609`, `eta.amp2609`, `pi0n.amp2609`
parameter file	`exclurad_py/models/amp2609_par.npy`, 37 long
md5	`abcb2f6ce7c87864386d900d82ad13ef`
origin	`~/rc_iter1/runs/C1_with_compass/fitpar.npy`
fitted to	`pi0_eta_database.xlsx`, md5 `8d5b6a04206de590a186b0f205eea44c`
quality	$\chi^2/\mathrm{ndf} = 1.4018 = 2140.5/1527$, eleven sets, 1555 points
computed by	`_amplitude_fit2.py` (the 37-parameter layout)

The installed vector was verified byte-for-byte against the fit it claims to come from. Three normalisations were fitted alongside it — 1.0586 for De Masi, 0.9776 for the Hall A neutron, 1.0583 for Hall A 2016 — and they are *not* applied here: they describe those experiments' normalisations, not the physics.

`amp2026c` is withdrawn. It holds the previous fit, `C0_with_compass`, which was made with a wrong kinematics for Hall A 2021: table I of PRL **127** 152301 heads its columns 0.36/0.48/0.60 and gives the per-setting $\langle x_B \rangle$ underneath, and the workbook had taken the heading, putting four of the nine settings at the wrong x_B by up to 0.03. Anything generated with `amp2026c` after 2026-09-05 21:01 carries that error. The practical size of it is small — between the two fits χ^2/ndf moves $1.3918 \to 1.4018$, $n_Q(H_T)$ $1.227 \to 1.260$, and cross sections shift by a few percent — so the events are not nonsense, but they were made by a model that no longer exists and should be labelled, not quietly kept. The name must not be reused: an installed name is the only link between a file on `wd_14tb` and the physics that made it.

2 What is sampled

The model supplies nine structure functions in nb/GeV^2. With an unpolarised target the generator samples samples

$$\frac{d^2\sigma}{dt\,d\phi} = \frac{1}{2\pi}\Big[\sigma_T + \epsilon\,\sigma_L + \epsilon\cos 2\phi\;\sigma_{TT} + \sqrt{2\epsilon(1+\epsilon)}\cos\phi\;\sigma_{LT} + h\sqrt{2\epsilon(1-\epsilon)}\sin\phi\;\sigma_{LT'}\Big], \quad (1)$$

with ϕ in the Trento convention and h the beam helicity. This is the decomposition the data were published in, so no conversion happens between the fit and the generator.

Since 2026-09-18 the generator also samples the four structure functions a *longitudinally polarised target* needs. With target polarisation Λ, set by `ptarg` in the card, the bracket gains

$$+\,\Lambda\Big(\sqrt{2\epsilon(1+\epsilon)}\sin\phi\;\sigma_{LT}^{UL} + \epsilon\sin 2\phi\;\sigma_{TT}^{UL}\Big) + \; h\Lambda\Big(\sqrt{1-\epsilon^2}\;\sigma_{T'} + \sqrt{2\epsilon(1-\epsilon)}\cos\phi\;\sigma_{LT'}^{LL}\Big), \quad (2)$$

the first line being A_{UL} and the second A_{LL}. All four are fitted: the eg1-dvcs moments of each are in the fit the parameters come from.

Three things about that are worth knowing.

The coefficients were measured, not assumed. Each candidate tensor structure was contracted with the lepton tensor over a grid in ϕ and its harmonic read off; the normalisation then follows from the same closure that fixes `NU_LT` and `NU_LTP` in `core/hadronic.py`, and that closure returns -1 for both of those, which is what makes the other four trustworthy. Born closure with Λ is flat to 10^{-8} over both helicities, $\Lambda \in \{0, \pm 1, 0.8\}$, nine ϕ and three kinematics.

The radiative correction is polarisation-dependent, correctly. `sigma_F` contracts the hadronic tensor at the *shifted* kinematics rather than scaling a scalar Born, so the polarised terms are integrated over photon emission like everything else, and the rotation of the photon frame against the fixed target spin falls out of the basis vectors being built from the actual momenta. η runs from 1.058 to 1.084 across (h, Λ) where before there was one number.

The spin axis is the virtual photon, not the beam. The two differ by the photon scattering angle, and the difference mixes a transverse target asymmetry into A_{UL}. The amplitude fit makes the same approximation, so generator and model agree and are approximate in the same way; the angle is the obvious next refinement, and it would require refitting eg1.

End to end, 8000 Born events at each of $\Lambda = \pm 0.8$ with the beam helicity fixed give $\langle 2\sin\phi\rangle = +0.248 \pm 0.017$ against -0.275 ± 0.018. The model gives $A_{UL}^{\sin\phi} = 0.30$ there, so the modulation should be $0.8 \times 0.30 = 0.24$.

Nothing else changed. The polarised terms are gated on both a non-zero Λ and a model that supplies the keys, and `ptarg` defaults to zero, so the vector-meson channels are bit-identical: born, η and $\sigma_{\rm obs}$ for `rho.diffrad_vm`, `phi.diffrad_vm`, `jpsi.diffrad_vm` and `jpsi.diffrad` were compared against a worktree at the previous commit over 48 values with zero differences.

3 Where it may be used

The model was fitted over

$$Q^2 \in [1.15,\ 8.31]\ \text{GeV}^2, \qquad x_B \in [0.13,\ 0.60], \qquad |t| \le 1.8\ \text{GeV}^2.$$

The production card `cards/rad_pi0_2026.txt` samples $W^2 \in [4, 12]$, $Q^2 \in [1, 5]$, $|t| \in [0.10,\ 2.0]$ at $E = 10.6$ GeV. Scanning that box on a grid of 62 900 kinematic points:

	points	share
inside the fitted region	50 000	79.5 %
$\lvert t\rvert > 1.8$	8 405	13.4 %
x_B outside $[0.13, 0.60]$	3 605	5.7 %
$Q^2 < 1.15$	3 198	5.1 %

So about a fifth of the sampled box is extrapolation, and the largest part of it is in $|t|$. Two remarks on that.

The card stops at $Q^2 = 5$ while the fit now reaches 8.31: the upper half of the fitted range is not being used at all. That is a card choice left from before Hall A 2021 entered the fit, and it is worth revisiting — that region is now measured rather than extrapolated.

Extrapolation in $|t|$ is the least safe of the three, because the t dependence is exactly what the form factors parametrise: eq. (1) of the report falls as $\exp[(b + b' \ln x_B)t]$ with no curvature, so beyond the data the shape is an assumption, not a measurement.

4 Positivity

A generator cannot sample from a negative weight, and nothing in the fit guarantees the bracket of eq. (1) stays positive: σ_{TT}, σ_{LT} and $\sigma_{LT'}$ enter with either sign and are not bounded by $\sigma_T + \epsilon\sigma_L$ in the parametrisation. It was therefore checked rather than assumed.

Over the card's box, on 263 808 points $(Q^2, W^2, |t|, \phi)$ with ϕ in 7.5° steps, the bracket never goes negative. Its smallest value relative to $\sigma_T + \epsilon\sigma_L$ is

$$\min \frac{[\,\cdot\,]}{\sigma_T + \epsilon\sigma_L} = 0.141 \qquad \text{at } Q^2 = 1.00,\ x_B = 0.178,\ -t = 2.00,\ \phi = 180^\circ,$$

which is at the far corner of the extrapolated region, not inside the data. The margin is comfortable, but it is a property of this vector: it must be re-checked whenever a new one is installed, and the scan above is cheap enough that there is no excuse not to.

5 The radiative-correction loop

This is the part most easily got wrong, so it is worth stating exactly.

Inside the generator the loop is closed by construction. `core/rc.py:rc_factor` takes a single `sfmodel` and passes the same object to `born_xsec` and to `sigma_F`, the integral over real photon emission. There is no second model inside the radiative correction: whatever the card names is used both for the Born cross section and for the soft and hard photon integrals, and

$$\eta = \sigma_{\rm obs}/\sigma_{\rm Born}$$

is computed with that one model throughout. Nothing needs fixing here.

On the data side the loop is at iteration one, and stopped there. The CLAS6 cross sections the model is fitted to are not the published ones: they were re-corrected by

$$\sigma_{\rm new} = \sigma_{\rm pub}/r, \qquad r = \frac{\eta(\text{new model})}{\eta(\text{published model})},$$

both η computed with `exclurad_py` so that the code cancels and only the model difference survives (`compute_rc.py`; published models `pi0.vpk2013` with $v_{\rm cut} = 0.18$ and `eta.tableB1` with $v_{\rm cut} = 0.094$).

The "new model" in that ratio is `amp2021`. The model that now describes those corrected data is `amp2609`, which is a different fit to a larger and repaired database. So the iteration that gives `rc_iter1` its name has been run once and not closed: the data carry the radiative correction of an August model, and the model fitted to them is a September one.

The size of the inconsistency is not known and is directly computable: recompute r with `amp2609`, apply it, refit, and see how far the structure functions and the parameters move. Until that is done, absolute rates from the generator are self-consistent *within* the generator but rest on data corrected with a different model. This is the one open item in the chain, and it is a real one.

6 What the generator inherits from the fit

These are measured failures of the model, not caveats added for form. Each is documented with numbers in the report; here is what each means for generated events.

σ_{LT} is the model's worst-described quantity. It carries the entire $\cos\phi$ modulation of eq. (1). Left out of the fit, the CLAS12 cross sections are predicted at 20.4 per point in σ_{LT} against 2.39 in σ_U and 3.51 in σ_{TT}; Hall A 2016 gives 4.7. The ϕ shape of the generated sample is therefore the least trustworthy thing it produces, and any analysis that leans on the $\cos\phi$ moment should be aware of it.

A single power of Q^2 spans 1.2 to 8.3, and does not fit. Hall A 2016 and Hall A 2021 disagree with the model in opposite directions at the two ends. With Hall A 2021 in the fit the compromise costs the asymmetries. Rates near the ends of the Q^2 range are less reliable than in the middle.

The neutron channel runs, and its normalisation is soft by a factor two. `pi0n.amp2609` works and is a genuine prediction, but n/p rests on four points with 42 to 90 % errors, and a single constrained variant of the fit — tying the t-slopes of $\bar{E}^u_T$ and $\bar{E}^d_T$, which costs 194 units of χ^2 — halves it, from 1.05 to 0.52, while the neutron data themselves fit *better*. Treat neutron output as indicative.

Two parameters sit on a bound. The t-slopes of H^u_T and of the longitudinal block are both exactly zero, held there by the requirement that form factors fall rather than grow with $|t|$. The constraint is deliberate and justified, but it means those slopes are limits, not measurements, and the extrapolation in $|t|$ noted above starts from a boundary value.

7 Reproducing a run

A production run is defined entirely by its card. For the record, the shape of one:

`pi0.amp2609`	model, and therefore the physics
`10.6`	beam energy, GeV
`200000`	events
`0`	`iborn`: 0 = full RC, 1 = Born only
`4.0 12.0`	W^2 range
`1.0 5.0`	Q^2 range
`-2.0 -0.10`	t range
`0.50`	v_{cut}
`0.80`	`ptarg`: longitudinal target polarisation, $-1 \ldots +1$
`20260828`	seed
`/Volumes/wd_14tb/mc/...`	output root

Run from `~/ampgen`, which holds the cards, the run scripts and the logs, with

```
python3 -m exclurad_py.run cards/<job>.txt --jobs 8
```

Two things about that directory are worth fixing. It is not under version control, so a card can change without leaving a trace, and the link between an output directory on `wd_14tb` and the card that produced it exists only by naming convention. For a generator whose whole value is reproducibility, that is the weakest link in the chain — weaker than anything in the physics above.

Where the rest is

the model and the fit	`~/rc_iter1/tex/report.pdf`, 28 pages
the radiative corrections	`doc/note/exclurad_py_note.pdf`
the formulas	`docs_src/rc_formulas.pdf`
using the code	`doc/manual/exclurad_py_manual.pdf`
installing a new vector	`doc/HANDOVER_amplitude_model_2026-09-05.md`
the fit, code and history	`https://github.com/vkubarovsky/rc_iter1`